UNIVERSITY OF ZIMBABWE

CAPSTONE PROJECT 2 PROGRESS ASSESSMENT 1

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VR VENUE SIMULATION TOOL FOR EVENT SETUP

Progress Assessment Table

Item No.	Software Module	Work Achieved	Possible Score	Actual Score
1	Data Collection	Secondary data collection was successfully completed. The project uses venue-planning requirements gathered from real event-planning workflows such as weddings, church services, conferences, exhibitions, funerals, and corporate events. The project also uses existing floor plan formats such as draw.io, XML, and HTML exports as input data sources for layout conversion.	6	
2	Data Processing	Data processing was successfully implemented. The backend processes uploaded draw.io, XML, and HTML floor plan files using XML parsing, shape extraction, label interpretation, geometry reading, grouped coordinate handling, and scaling from measurement labels. The processed data is converted into editable venue scene objects. Tools used include Django, Django REST Framework, Python XML parsing, and custom layout import services.	8	
3	AI Layout Generation Module	The AI layout generation module was built. The system accepts a natural language prompt, sends it to a local Ollama Gemma 3 model through the Django backend, validates the generated layout intent, and converts it into deterministic venue objects. The model output is	8	

		controlled using structured schemas instead of relying on raw free-form AI coordinates.		
4	AI Model Performance Evaluation	Initial AI performance evaluation was completed through functional prompt testing. The model successfully generated structured layout intent for banquet, theatre, classroom, boardroom, lounge, and exhibition layouts. Evaluation focused on layout accuracy, room size compliance, valid object categories, and scene usability. Further quantitative evaluation metrics are still to be expanded.	5	
5	Login Module	The login module was built and tested. Users can log in using the backend authentication API. JWT authentication is handled using HttpOnly cookies for improved security. Test scenarios included valid login, invalid login, protected page access after login, and logout flow.	6	
6	Signup / Registration Module	The signup module was built and tested. New users can create accounts through the frontend registration page, and the backend validates and stores the new user account. Test scenarios included successful registration and prevention of invalid registration data.	6	
7	Protected Dashboard Module	The protected dashboard module was built. Authenticated users can view and manage their own projects. The dashboard is connected to the project API so that each user only accesses projects belonging to their account.	6	

8	Project Management Module	The project management module was implemented. Users can create, save, update, and retrieve venue projects. Each project stores room dimensions, scene items, AV connections, measurements, scene settings, creation time, and update time using the Django backend and SQLite database.	8	
9	2D Venue Editor Module	The 2D editor module was built. Users can view the venue floor plan, drag objects, resize the room, select items, use snapping, and prevent object placement outside the room. The editor includes collision checking, room clamping, alignment guides, and layer filtering for layout and AV items.	10	
10	3D Venue Visualization Module	The 3D scene module was built using React Three Fiber, Drei, and Three.js. Users can inspect the room in 3D with placed objects such as chairs, tables, podiums, screens, cameras, speakers, altar, desk, piano, and other venue assets.	8	
11	Asset Catalog Module	The asset catalog module was implemented using integration with the Poly Pizza API for sourcing 3D venue and AV assets. The system retrieves asset models such as chairs, banquet tables, desks, podiums, screens, TVs, altars, cameras, speakers, pianos, church benches, and other venue objects from the external asset service, then caches and stores the asset metadata in the Django backend for faster access and improved reliability. This allows users to load assets efficiently inside the 2D/3D venue editor, place them within layouts, and edit them in the scene without repeated external API calls.	5	

12	AV Planning and Cable Connection Module	The AV planning module was built. The system supports AV equipment placement, cable connection logic, cable type inference, and visible connections between compatible devices such as cameras, screens, speakers, and mixing desks.	6	
13	Floor Plan Import Module	The floor plan import module was implemented. Users can import draw.io, XML, and HTML floor plan files. The backend extracts shapes, labels, dimensions, grouped objects, and object types, then converts them into editable project scenes.	6	
14	Templates Module	The template module was implemented. Users can start from prebuilt venue templates, reducing setup time and helping non-technical users begin quickly. Templates are converted into editable project copies.	3	
15	Sharing and Export Module	The sharing and export module was built. Users can generate shareable read-only project links and export 2D floor plans as PNG or PDF files for presentation and client review.	4	
16	Presentation Skills	The project can be demonstrated through registration/login, dashboard, project creation, AI prompt generation, imported floor plan conversion, 2D editing, 3D visualization, AV cable planning, and export/share features.	5	
	TOTAL		**100**	

Not yet achieved Modules

Item No.	Feature Yet To Come	Current Status	Planned Improvement
1	Manual wall drawing tools	Not Achieved	Allow users to draw, move, resize, and delete walls directly inside the editor
2	Manual cable routing with bend points	Not Achieved	Allow AV users to manually adjust cable paths instead of only using automatic cable routes
3	Project versioning	Not Achieved	Add restore points so users can recover previous versions of a venue design
4	First-Person Walkthrough Mode	Not Achieved	Add an interactive first-person walkthrough mode using keyboard and mouse controls so users and clients can navigate inside the 3D venue space.
5	Accessibility and Safety Checking	Not Achieved	Add automatic checks for aisle spacing, overcrowding, walking paths, emergency access, and object collision risks. The system should provide warnings when layouts do not meet safe movement or accessibility requirements.
6	Printable Project Summary	Not Achieved	Extend the export module to generate a printable project summary containing room dimensions, layout type, item counts, seating capacity, equipment list, and project details for client review, quotations, and assessment documentation.
7	Equipment Bill of Material	Not Achieved	Add an automatic equipment count and bill of materials showing the number of chairs, tables, speakers, cameras, screens, podiums, cables, and other venue

			assets required for the event. This improves logistics planning and quotation preparation.

Assessor Section

ASSESSOR'S SIGNATURE: ____

DATE: ____